Student ID _____ Student Name _____

IS222: Database Management Systems

Faculty of Science, Technology and Environment
School of Computing, Information and Mathematical Sciences

Final Examination
Semester I 2019

Face-To-Face and Online Mode

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 180 minutes

INSTRUCTIONS

1. This examination has four sections:
   - Section A: Multiple Choice (20 marks)
   - Section B: Modelling & Normalization (40 marks)
   - Section C: SQL (25 marks)
   - Section D: Transaction (15 marks)
2. **All** questions are compulsory.
3. Write **ALL** answers in the space provided within this paper.
4. This exam is **marked out of 100** and is worth **40%** of your **overall course marks**.
5. To **pass** this paper you require **at least 40% for this exam**, which is equivalent to **40 marks out of 100.**
6. This is a closed book exam.
7. This paper has 23 pages including this page.

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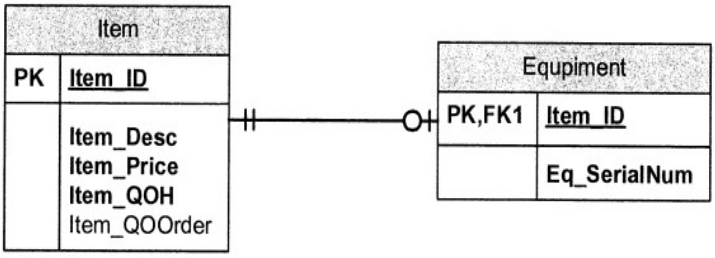
Section A: Multiple Choice 20 Marks

Instructions:
Identify and circle the choice that best completes the statement or answers the question in the provided answer sheet on page 8. There is only **one** possible answer.

1. What serves as an intermediary between the user and the database:
   - **a. DBMS**
   - b. Metadata
   - c. End-user data
   - d. Programming language

1) Which of the following is incorrect about business rules (BR)?
   - a. A BR is bidirectional
   - **b. A BR states the cardinality of a relationship**
   - c. A BR states connectivity of a relationship
   - d. A BR describes the relationship between two entities

2) Which of the following best describes the properties of the relationship shown below?



   - a. Weak, Optional
   - **b. Strong, Optional**
   - c. Weak, Mandatory
   - d. Strong, Mandatory

3) What does NoSQL mean?
   - a. Not on SQL
   - b. Not in SQL
   - **c. Not only SQL**
   - d. Note on SQL

4) Which of the following can be null?
    a. Primary Key
    b. Foreign Key
    c. Both (A) and (B) can be null
    d. Both (A) and (B) cannot be null

5) Which of the following is the same as a Row in a table?
    a. Attribute
    b. Tuple
    c. Relation
    d. Data

6) The _____ model is the end users' view of the data environment.
    a. Abstract
    b. External
    c. Conceptual
    d. Internal

7) Which key is solely responsible for data retrieval?
    a. Foreign Key
    b. Primary Key
    c. Super Key
    d. Secondary Key

8) The student table with a primary key STU_ID should not have which of the following dependency?
    a. Functional
    b. Partial
    c. Transitive
    d. All of the above

9) Given below is the ERD for the Patient table. Which attribute is a suitable candidate key

Patient	
PK	Pat_Code
	Pat_LName Pat_FName Pat_DOB Pat_Addr Pat_Phone

a. Pat_Lname
b. Pat_DOB
c. Pat_Addr
d. Pat_Phone

10) Which of these data types in MySql can be used to represent the value, AB23?

a. VCHAR
b. CHAR
c. STRCHAR
d. None of the above

11) To delete a row from a table, use the ____ command.

a. COMMIT
b. DELETE
c. DROP
d. ERASE

12) The ____ statement in SQL combines rows from two queries and returns only the rows that appear in the first set but not in the second.

a. UNION
b. UNION ALL
c. INTERSECT
d. **MINUS**

13) As long as two transactions, T1 and T2, access ____data, there is no conflict, and the order of execution is irrelevant to the outcome.

a. shared
b. common
c. **unrelated**
d. locked

14) One of the three most common data integrity and consistency problems is ____.

a. user errors
b. **lost updates**
c. disk failures
d. deadlocks

15) A consistent database is ____.

a. one in which all tables have foreign keys
b. one in which all tables are normalized
c. one in which all SQL statements only update one table at a time
d. **one in which all data integrity constraints are satisfied**

16) Granularity refers to ____.

a. the size of a table
b. the number of attributes in a table
c. the number of rows in a table
d. **the level of detail represented by the values stored in a table's row**

17) How many rows would be returned from a cross join of tables A and B, if A contains 8 rows and B contains 18?

a. **144**
b. 26
c. 18
d. 8

18) ____ are required to prevent another transaction from reading inconsistent

a. **Locks**
b. Stamps
c. Logs
d. Schedules

19) A(n) ____ is an alternate name given to a column or table in any SQL statement.

a. data type
b. stored function
c. alias
d. trigger

20) Which of the following SQL statements will correctly create an index on the stud_lname field of the student table in a MySQL database.

a. `CREATE INDEX stud_lname ON student;`
b. `CREATE INDEX stud_idx_stud_lname ON stud_lname (student) ;`
c. `CREATE INDEX stud_idx studl_name FROM student (stud_lname) ;`
d. **`CREATE INDEX stud_idx_stud_lname ON student (stud_lname);`**

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Section A: Answer Sheet

Circle the correct answer.

1.	**a**	**b**	**c**	**d**
2.	**a**	**b**	**c**	**d**
3.	**a**	**b**	**c**	**d**
4.	**a**	**b**	**c**	**d**
5.	**a**	**b**	**c**	**d**
6.	**a**	**b**	**c**	**d**
7.	**a**	**b**	**c**	**d**
8.	**a**	**b**	**c**	**d**
9.	**a**	**b**	**c**	**d**
10.	**a**	**b**	**c**	**d**
11.	**a**	**b**	**c**	**d**
12.	**a**	**b**	**c**	**d**
13.	**a**	**b**	**c**	**d**
14.	**a**	**b**	**c**	**d**
15.	**a**	**b**	**c**	**d**
16.	**a**	**b**	**c**	**d**
17.	**a**	**b**	**c**	**d**
18.	**a**	**b**	**c**	**d**
19.	**a**	**b**	**c**	**d**
20.	**a**	**b**	**c**	**d**

Section B: Modeling & Normalization 40 Marks

a) Give an example of a computerized file system and list any three of its major limitations? **(4 marks)**

Examples - Text files on computers, excel files

Limitations -

1. Lengthy development times
2. Difficult to manage and maintain
3. Lack of security
4. Difficulty in getting quick answers
5. Complex administration
6. Requires extensive programming

b) What is structural dependence in the context of a file system **(2 marks)**

A data characteristic in which a change in the database schema affects data access, thus requiring change in all access programs

c) List and explain the insert and update anomalies? **(3 marks)**

Insert Anomaly – if a tuple is inserted in referencing relation and referencing attribute value is not present in referenced attribute, it will not allow inserting in referencing relation. For Example, If we try to insert a record in STUDENT_COURSE with STUD_NO =7, it will not allow.

Update Anomaly - If a tuple is deleted or updated from referenced relation and referenced attribute value is used by referencing attribute in referencing relation, it will not allow deleting the tuple from referenced relation. For Example, If we try to delete a record from STUDENT with STUD_NO =1, it will not allow.

d) How do you decide whether a table has referential integrity? **(2 marks)**

In the context of relational databases, it requires that if a value of one attribute (column) of a relation (table) references a value of another attribute (either in the same or a different relation), then the referenced value must exist.

e) Why are keys needed in a database? **(2 marks)**

Keys are very important part of Relational database model. They are used to establish and identify relationships between tables and also to uniquely identify any record or row of data inside a table.

f) List and explain any three features of a DBMS **(3 marks)**

Insulation between program and data - In the file-based system, the structure of the data files is defined in the application programs so if a user wants to change the structure of a file, all the programs that access that file might need to be changed as well.

Support for multiple views of data - A database supports multiple views of data. A view is a subset of the database, which is defined and dedicated for particular users of the system. Multiple users in the system might have different views of the system. Each view might contain only the data of interest to a user or group of users.

Control of data redundancy - In the database approach, ideally, each data item is stored in only one place in the database.

Backup and recovery facilities - Backup and recovery are methods that allow you to protect your data from loss. The database system provides a separate process, from that of a network backup, for backing up and recovering data.

Restriction of unauthorized access - Not all users of a database system will have the same accessing privileges, should provide a security subsystem to create and control different types of user accounts and restrict unauthorized access

Functions (any 3 would do) -> data dictionary management, data storage management, data transformation and presentation, security management, multiuser access control, backup and recovery management, data integrity management, database access languages and application programming interfaces, database communication ..

g) How do you decide if an entity is weak or strong? Given an example of such an entity **(3 marks)**

An entity is considered weak if its tables are existence dependent. The Spouse table, in the COMPANY database, is a weak entity because its primary key is dependent on the Employee table. Without a corresponding employee record, the spouse record would not exist.

h) What is a derived attribute? Give relevant examples to support your answer **(3 marks)**

A derived attribute is an attribute whose value is calculated (derived) from other attributes. The derived attribute need not be physically stored within the database; instead, it can be derived by using an algorithm. For example, an employee's age, EMP_AGE, may be found by computing the integer value of the difference between the current date and the birth date of the employee.

i) Discuss the difference between a composite key and a composite attribute. **(2 marks)**

A composite key is one that consists of more than one attribute. If the ER diagram contains the attribute names for each of its entities, a composite key is indicated in the ER diagram by the fact that more than one attribute name is underlined to indicate its participation in the primary key.

A composite attribute is one that can be subdivided to yield meaningful attributes for each of its components. For example, the composite attribute CUS_NAME can be subdivided to yield the CUS_FNAME, CUS_INITIAL, and CUS_LNAME attributes. There is no ER convention that enables us to indicate that an attribute is a composite attribute

j) Given the following business scenario, create the appropriate business rule (s). **(2 marks)**

Samsung has recently released their galaxy series 10 devices. Each Galaxy phone has its own specifications which includes details such as the storage capacity, processor speed, screen resolution and others. The phones are being sold at various retail stores across Fiji such as Vodafone, Digicel, Bondwell and others. In order to attract maximum customers, all retail stores sell a variety of phones.

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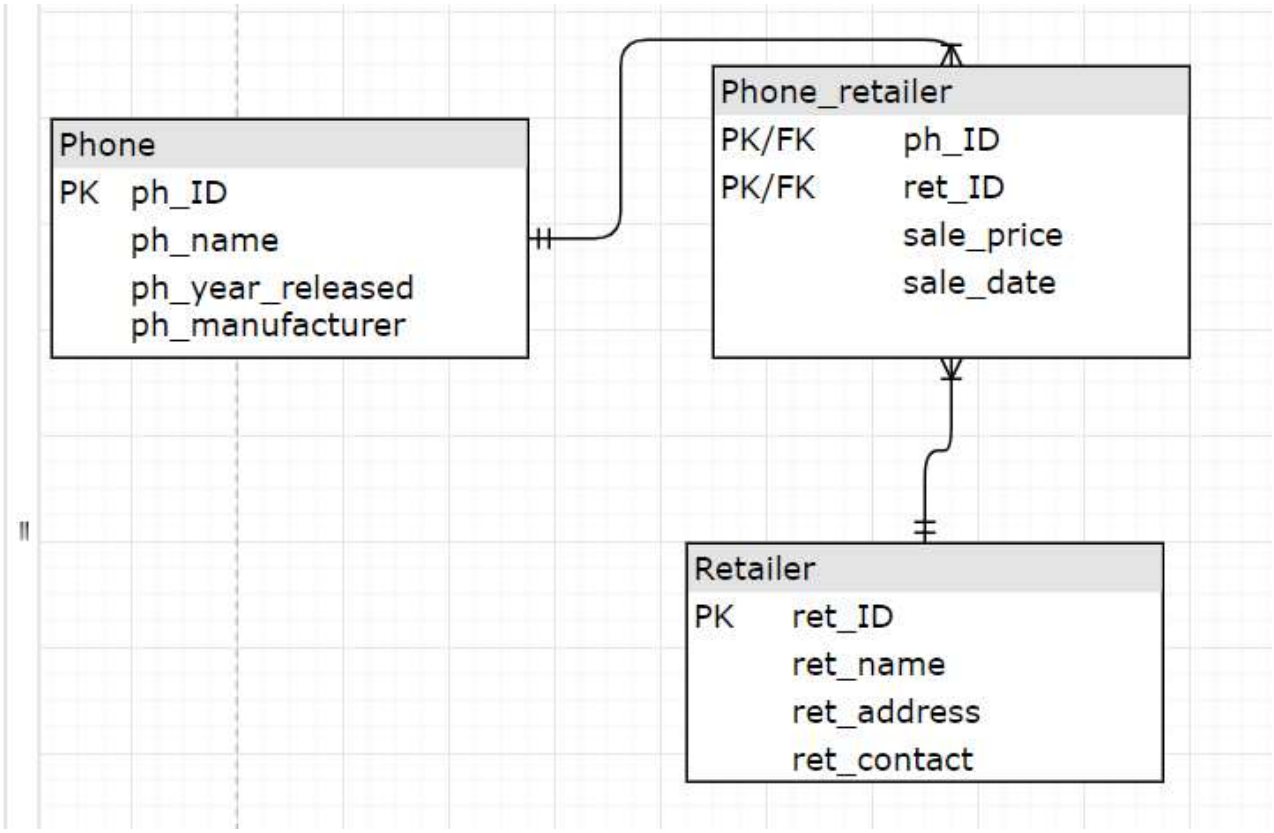
1. A phone is sold at many retail stores and a retail store can sell many phones.

k) Draw an Entity Relationship Diagram to depict the business rules mentioned above. You must include some attributes that you think should be part of each entity. **(4 Marks)**



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l) What is the purpose of normalization? **(2 marks)**

Database normalization is the process of structuring a relational database in accordance with a series of so-called normal forms in order to reduce data redundancy and improve data integrity..

m) Normalize the given table. You must show all the intermediary steps. **(8 Marks)**

user_orders								
order_id	**sku**	**name**	**line_total**	**order_total**	**order_date**	**user_id**	**user_joined**	**user_address_state**
12345	S-12	Blue shirt	10.50	41.45	2015-01-01	123	2014-12-30	Delaware, New Jersey
12345	J-34	Windbreaker	23.00	41.45	2015-01-01	123	2014-12-30	Delaware, New Jersey
12345	B-13	Belt	7.95	41.45	2015-01-01	123	2014-12-30	Delaware, New Jersey
67891	J-35	Denim Jacket	21.00	36.75	2015-01-02	456	2014-12-31	Pennsylvania
67891	H-78	White Hat	15.75	36.75	2015-01-02	456	2014-12-31	Pennsylvania
111213	S-12	Blue shirt	10.50	28.50	2015-01-03	123	2014-12-30	Delaware, New Jersey
111213	H-78	White Hat	18.00	28.50	2015-01-03	123	2014-12-30	Delaware, New Jersey

Order_id, sku -> name, line_total, order_total, order_date, user_id, User_joined, user_address

Sku -> name, line_total (PD)

Order_id -> order_total, order_date, user_id, User_joined, user_address (PD)

user_id -> User_joined, user_address (TD)

Order_id, sku (Original table)

Section C: Structured Query Language (SQL) 25 Marks

Question 1

What are "referential constraint actions"? **(2 marks)**

Referential constraint actions, such as ON DELETE CASCADE, are default actions that the DBMS should take when a DML command would result in a referential integrity constraint violation. Without referential constraint actions, DML commands that would result in a violation of referential integrity will fail with an error indicating that the referential integrity constrain cannot be violated. Referential constraint actions can allow the DML command to successfully complete while making the designated changes to the related records to maintain referential integrity.

Question 2

Explain why the following command would create an error, and what changes could be made to fix the error. **(2 marks)**

```
SELECT V_CODE, SUM(P_QOH) FROM PRODUCT;
```

The command would generate an error because an aggregate function is applied to the P_QOH attribute but V_CODE is neither in an aggregate function or in a GROUP BY.

This can be fixed by either

1) placing V_CODE in an appropriate aggregate function based on the data that is being requested by the user,

2) adding a GROUP BY clause to group by values of V_CODE (i.e. GROUP BY V_CODE),

3) removing the V_CODE attribute from the SELECT clause, or

4) removing the Sum aggregate function from P_QOH. Which of these solutions is most appropriate depends on the question that the query was intended to answer

Question 3

Suppose that you have two tables, EMPLOYEE and EMPLOYEE_1. The EMPLOYEE table contains the records for three employees: Alice Cordoza, John Cretchakov, and Anne McDonald. The EMPLOYEE_1 table contains the records for employees John Cretchakov and Mary Chen. Given that information, what is the query output for the UNION query? (List the query output.)

(2 marks)

The query output will be:

Alice Cordoza
John Cretchakov
Anne McDonald
Mary Chen

Question 4

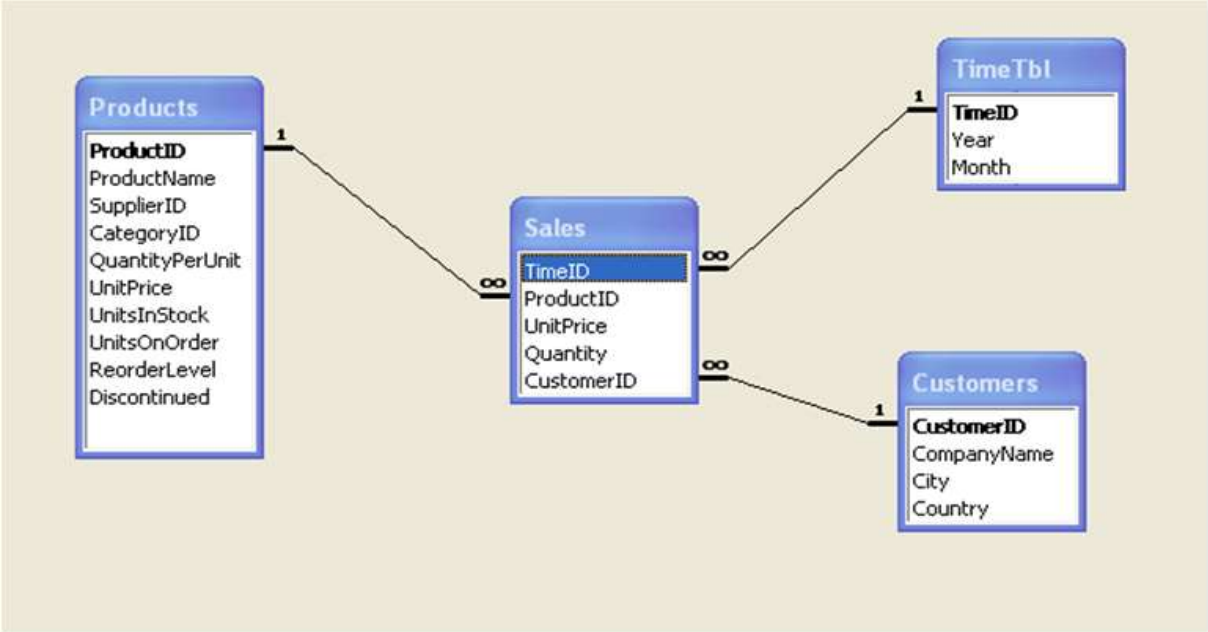
What is query optimization? **(1 mark)**

It's a technique used by a relational database management system to find the efficient access path to a data when executing a query

Question 5

Assume the following database is used to store the sales data for a company. Use the following description and the database schema given below to answer the following questions of this section.

The management team needs some reports for their upcoming meeting. In the following questions, you will be required to write the relevant queries that will provide the required information for their report. Your queries should be written for a MySQL database. You are required to use the given attribute names in the following ERD diagram in your queries.



a) A user named Sally needs to access the database. Write the necessary SQL command(s) so that Sally can login to her MySql account given that her username is Sally and password is Secret. **(2 marks)**

```
mysql -u Sally -pSecret;
```

b) Based on your experience gained from the IS222 lab exercises, Write the MySQL command that will create a database named Sales. **(1 mark)**

```
CREATE DATABASE Sales;
```

c) The structure of the Customers table is given in Figure 1 below. Write the necessary SQL Statement to create the Customers table **(4 Marks)**

Field	Type	Null	Key	Default
CustomerID	varchar(5)	NO	PRI	*NULL*
CompanyName	varchar(40)	NO	MUL	*NULL*
City	varchar(15)	YES	MUL	*NULL*
Country	varchar(15)	YES		*NULL*

Figure 1

```
--
-- Table structure for table `customers`
--
DROP TABLE IF EXISTS `customers`;
CREATE TABLE IF NOT EXISTS `customers` (
  `CustomerID` varchar(5) NOT NULL,
  `CompanyName` varchar(40) NOT NULL,
  `City` varchar(15) default NULL,
  `Country` varchar(15) default NULL,
  PRIMARY KEY  (`CustomerID`),
  INDEX `City` (`City`),
  INDEX `CompanyName` (`CompanyName`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8;
```

d) List the country and the number of customers from that particular country

(1 mark)

```
Select Country , count(CustomerID)
From Customers
Groupby Country
```

e) List the number and name of product whose category is 2, unit price is greater than 18 and product name not starting with “Chef”. **(2 marks)**

```
SELECT ProductID, ProductName

FROM products

WHERE `CategoryID` =2 AND `UnitPrice` >=18 AND `ProductName` NOT LIKE "Chef%";
```

f) List the number(s), name(s) and price of the most and least expensive product(s) **(3 marks)**

```
Select ProductID, ProductName, UnitPrice
from products
where UnitPrice = (select max(UnitPrice) from products)
UNION
Select ProductID, ProductName, UnitPrice
from products
where UnitPrice = (select min(UnitPrice) from products);

OR

Select ProductID, ProductName, UnitPrice
from products
where UnitPrice = (select max(UnitPrice) from products) or UnitPrice = (select
min(UnitPrice) from products);
```

g) Update UnitsInStock to zero for whose product id is 33? **(2 marks)**

```
Update products
Set UnitsInStock = 0
Where ProductID = 33;
```

h) Add a column named “sum” whose data type is integer in products table. Update sum by adding the UnitsInStock and UnitsOnOrder whose supplier id and category id are the same. **(3 marks)**

```
Alter table products
add (sum Integer);
UPDATE products
SET sum = UnitsInStock + UnitsOnOrder
WHERE SupplierID = CategoryID;
```

Section D: Transaction Processing 15 Marks

Question 1

What is transaction isolation? **(2 marks)**

Isolation means that the data used during the execution of a transaction cannot be used by a second transaction until the first one is completed. In other words, if transaction T1 is being executed and is using the data item X, that data item cannot be accessed by any other transaction (T2 ... Tn) until T1 ends.

Question 2

What is the purpose of a transaction log? **(2 marks)**

The transaction log is a special DBMS table that contains a description of all the database transactions executed by the DBMS. The database transaction **log plays a crucial role in maintaining database concurrency control and integrity**.

The information stored in the log is used by the DBMS to recover the database after a transaction is aborted or after a system failure. The transaction log is usually stored in a different hard disk or in a different media (tape) to prevent the failure caused by a media error.

A DBMS uses a transaction log to **keep track of all transactions** that update the database. The DBMS uses the information stored in this log **for a recovery requirement** triggered by a ROLLBACK statement, a program's abnormal termination, or a system failure such as a network discrepancy or a disk crash.

Question 3

List and discuss the five transaction properties. **(5 marks)**

The five transaction properties are:

Atomicity requires that all parts of a transaction must be completed or the transaction is aborted. This property ensures that the database will remain in a consistent state.

Consistency Indicates the permanence of the database consistent state.

Isolation means that the data required by an executing transaction cannot be accessed by any other transaction until the first transaction finishes. This property ensures data consistency for concurrently executing transactions.

Durability indicates that the database will be in a permanent consistent state after the execution of a transaction. In other words, once a consistent state is reached, it cannot be lost .

Serializability means that a series of concurrent transactions will yield the same result as if they were executed one after another.

Question 4

Consider the two transactions, T1 and T2, given below

```
T1:  Read1(A)
     Read1(B);
     If(A==0) Then B = B + 1
     Write1(B);

T2:  Read2(B);
     Read2(A);
     If (B==0) Then A = A + 1
     Write2(A);
```

Read1 and Read2 are read operations for their respective transactions, 1 and 2.

Write1 and Write2 are write operations for their respective transactions, 1 and 2.

Assume that both A & B are 0 initially with either A=0 or B=0. Answer the questions that follow in the spaces given:

a) Will serial execution of these transactions always preserve consistency (yes/no)? **(1 mark)**

Ans: Yes

b) Explain your answer in part (a) above. **(1 mark)**

Marker note:

Students may explain differently and can get partials depending on the accuracy of explanation. You have to read to make sense of the explanation. **Apply discretion**

Answer has to consider two possible cases of schedules.

- **1st case:** T1 executes and then T2, will mean *A=0 and B=1 at the end of execution, so consistency is preserved*
- **2nd case:** T2 executes and then T1, will mean *A=1 and B=0, at the end of execution, so consistency is preserved*

In both cases, consistency is preserved

c) Identify all conflicting operations during concurrent execution of the transactions**.** **(1 mark)**

Ans

- Read1(A), Write2(A) **(0.5 mk)**

- Read2(B), Write1(B) **(0.5 mk)**

Question 5

Consider that there are only two transactions, T1 and T2, with timestamps TS1 and TS2, respectively as shown in table 1 below. X, Y and Z are data items. Assume that T2 first requests for exclusive lock on Y, at the same time, T1 requests for exclusive lock on X. Then T1 next requests a lock on Y.

Transaction T2 TS2 "2019-04-15 22:50:56"	**Transaction T1 TS1** "2019-04-15 22:51:12"
	Read (X)
	Read (Y)
Read (Y)	
Read (Z)	
	Write (X)
Write (Y)	
Write (Z)	

Table 1

Explain specifically how the locks will be acquired/released by the transactions from the outset and how the transactions will come to completion using the Wait-die algorithm. **(3 marks)**

Ans:

- First: T2 gets a lock on Y and T1 gets a lock on X
- Next, T1 requests a lock on Y which cannot be granted as the lock is held by T2. By the wait-Die scheme, T1 will die/rollback and is rescheduled using the same timestamp - T2 will acquire the locks on Z and complete, releasing all its locks when done.

Marker Note

This is how the tasks will happen from the outset. You can give marks based on the extent of correctness of the points. Apply discretion where necessary. Read the answer carefully of the student.

Full marks if everything is correct.

THE END

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